Raj Jung Mahat

Ph.D. Candidate, Mechanical Engineering

Computational solid mechanics · Bayesian inverse problems · UQ

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Summary

Ph.D. candidate at Georgia Tech (advisor Prof. Surya Kalidindi; doctoral minor in Mathematics; expected 2027). I build nonlinear finite-element models of mechanical tests and use them to infer material laws from measurements, with uncertainty. The work has three strands: (i) FE modeling of contact-dominated tests (small punch, indentation, tension) with plasticity, kinematic hardening and shape-memory constitutive models, in Abaqus (including UMATs) and in a differentiable FE code I wrote in JAX; (ii) Bayesian calibration with Gaussian-process surrogates, MCMC and model-discrepancy modeling, validated across nine alloys; (iii) the mathematics of what a test can identify: sensitivity operators, Fisher information and identifiability, and Gaussian measures for SPDE priors. One semester in industry as a data scientist (Multiscale AI).

Core skills

FE & mechanics	Nonlinear FE (finite strain, J_2 plasticity, return mapping, consistent tangents, contact); Abaqus/Standard incl. UMAT development and debugging (Fortran); constitutive models: Voce isotropic, Armstrong–Frederick/Chaboche kinematic, Lagoudas-type shape-memory; mesh and contact verification against Abaqus
Inference & UQ	Bayesian calibration, Gaussian-process surrogates on PCA-reduced outputs, MCMC (transitional, Hamiltonian, Riemannian-manifold), normalizing flows, model discrepancy, adjoint and forward-mode sensitivities, Fisher information, likelihood-informed subspaces, experimental design
Statistics & data	Survival analysis (Cox proportional hazards, censoring), multivariate statistics, design of experiments, PCA and diffusion maps; composition/processing–property data for alloy campaigns
Mathematics	Numerical linear algebra, numerical methods for PDEs/ODEs, Gaussian measures on function spaces (Feldman–Hájek, SPDE/Matérn priors), generalized eigenproblems
Software	Python (JAX, NumPy, SciPy, PyTorch, Pyro, PyMC, scikit-learn, pandas, lifelines), Fortran, MATLAB, R, SQL, LaTeX; Git, Linux, Slurm batch jobs on Georgia Tech's PACE Phoenix cluster

Research experience

2022–present **Graduate Research Assistant**, *Georgia Institute of Technology*, Atlanta, GA

Advisor: Prof. Surya Kalidindi. Constitutive-law inference from high-throughput mechanical tests.

Bayesian calibration from small punch tests (dissertation core; first-author manuscript in preparation)

- Built the pipeline from FE simulation to posterior: Abaqus small-punch and tensile models, PCA of the load–displacement response (5–7 components), Gaussian-process emulators (ARD kernels), and transitional and Hamiltonian MCMC, so the likelihood needs no FE solve. Applied to nine alloys (Al 2024/5052/6061/7075, Cu, brass, Inconel 718, nickel silver, SS316) and benchmarked against tensile tests.
- Model discrepancy: one fixed direction captures 93.7% of the punch–tensile mismatch; a response-dependent correction in latent PC space reduced cross-test error from about 28% to 14%.

Differentiable FE in JAX and the information content of a test (manuscript in preparation)

- Wrote an axisymmetric finite-strain FE code for the small punch test in core JAX (J_2 plasticity, implicit return map, penalty contact, load stepping), verified against Abaqus on identical meshes.
- Exact sensitivities of the force record to the full hardening curve by forward mode, adjoint and a hand-derived recursion (agreement to 10^{-13}). From them, the information operator of a test on the hardening function.
- Findings: across nine metals the punch places 90% of its information below 50–60% of the largest strain it reaches (tension: 88–109%). The FE-predicted weakest parameter direction matches the sampled posterior to 0.3–7° for five metals and misses for three, the same three where the emulator fails validation. Over 17 test geometries, resolved information ranks the benefit of a second test (rank correlation 0.86–0.89); reached strain and stress-state variety do not.

Kinematic hardening and shape-memory alloys (ongoing)

- Combined Voce + Chaboche hardening in the JAX solver, validated against Abaqus' combined-hardening model; which backstress components force, tensile and DIC data can separate.
- NiTi: re-implemented a Lagoudas-type finite-strain shape-memory model in JAX from the published equations (log strain with a custom derivative exact at repeated eigenvalues); it matches the Fortran UMAT to

4×10^{-3} MPa. Diagnosed the existing UMAT's failures in 3D (transformation-branch selection); Abaqus punch runs on the HPC cluster.

High-throughput alloy campaigns (HTMDEC / BIRDSHOT, FCC high-entropy alloys)

- Small punch, tensile, EBSD and composition data across eight vacuum-arc-melted batches; simulated about 60 HEAs and linked the systematic punch simulation–experiment mismatch to grain size (co-author, *Scripta Mater.* 2026).
- Population prior on hardening curves from composition and processing (349 tensile curves); leave-one-alloy-out spread of the predicted flow stress at strain 0.8 reduced from 24.5% to 14.6% (140 alloys).

Experiments, mentoring, teaching

- Designed a camera-instrumented (DIC) small punch rig for full-field strain; contributed experiments to a punch + DIC amortized-inference study (under review). Mentored undergraduates on DIC; TA, *Machine Learning for Mechanical Engineers* (Spring 2024).

2018–2021 **Graduate Research Student**, *Indian Institute of Science (IISc)*, Bangalore, India

Uniaxial creep properties from instrumented indentation: FE-generated training data and neural-network inverse models mapping the indentation response to creep parameters.

Industry experience

Jan–May 2025 **Data Scientist**, *Multiscale AI*, Atlanta, GA

- Designed and implemented modules of the company's ML platform (inference workflow, data workflow, feature-engineering pipeline) with a logging backend for run traceability.
- Built an inverse-inference module (surrogate design, MCMC sampling) packaged for reuse across client problems.
- Cox proportional-hazards model of adhesive-bond durability for a Novelis aluminum surface-treatment study: time to failure under humidity against laser-treatment parameters and surface measurements (roughness, oxide ratio), with right-censored observations.

Independent work

SPDE priors Sole-author preprint: when covariance or differential-operator changes to an SPDE prior give equivalent versus singular Gaussian measures, and finite-dimensional diagnostics under refinement for replicated fields. Open notes and notebooks: rajjung421.github.io/gaussian-measure-equivalence.

Sampling Leapfrog HMC from scratch with a study of acceptance and mixing versus step size and trajectory length on ODE-defined posteriors; Riemannian-manifold MCMC and normalizing flows benchmarked against HMC.

Education

2022–2027 (exp.) **Ph.D., Mechanical Engineering**, *Georgia Institute of Technology*, Atlanta, GA

Doctoral minor: Mathematics. Dissertation: Bayesian constitutive calibration and identifiability under compressed mechanical observation. Coursework: Finite Element Method, Micromechanics, Numerical Linear Algebra, Numerical Methods for PDEs and ODEs, Multivariate Statistics, Statistical Methods, Advanced Design of Experiments, Computational Data Analysis.

2018–2021 **M.S., Materials Engineering**, *Indian Institute of Science*, Bangalore, India

2013–2017 **B.E., Mechanical Engineering**, *M.S. Ramaiah Institute of Technology*, Bangalore, India

Publications and manuscripts

2026 **R. J. Mahat**. Measure equivalence as a guide for learning SPDE priors from replicated fields. Preprint. doi; alphaXiv

under review S. Mahmoud, A. Venkatraman, **R. J. Mahat**, et al., S. R. Kalidindi. Amortized posteriors for estimation of material constitutive parameters from multimodal measurements on small punch tests. arXiv:2607.24534.

2026 N. Flores, et al. (incl. **R. Mahat**, S. R. Kalidindi, R. Arróyave). Data-driven insights into composition–property relationships in FCC high entropy alloys. *Scripta Materialia* 274, 117136. doi

2024 M. Mulukutla, A. N. Person, S. Voigt, et al. Illustrating an effective workflow for accelerated materials discovery. *Integrating Materials and Manufacturing Innovation* 13, 453–473. doi

in prep. **R. J. Mahat**, A. Venkatraman, S. R. Kalidindi. Bayesian calibration of Voce hardening parameters from the small punch test: identifiability and structured cross-test discrepancy.

in prep. **R. J. Mahat**, et al. What a mechanical test resolves about a hardening function, and why a constitutive family turns it into a parameter ridge.